

SCIENCE NOTES YEAR 9

Biology: Genes and Inheritance

Chromosomes, Genes and DNA

The nucleus of the cell is larger than one of an atom.

The nucleus of both a cell and an atom is the centre of a large structure

Chromosomes= long threads of DNA, in nucleus and made from genes

DNA= a chemical substance that has a code of instructions that determines how an organism/cells develops

Genes= instructions in the form of DNA, affects characteristics

There are 23 pairs of chromosomes(total 46)

Genes makes up chromosomes and DNA makes up genes

If a human has an X and Y chromosomes= boy

If a human has 2 Y chromosomes= girl

Chromosomes are in a cross shape with four arms, this is how they look like when they make a copy of themselves

Chromosomes are normally twisted and not in a straight line

Diagram:

A pair of chromosomes are called chromatids and one chromosome is called a chromatids

They are joined together by their centromere

A chromosome made up from coiled up DNA

DNA is twisted into a double helix

DNA is made up from a sequence bases (A,C,T G) only specific bases can be paired up(complementary base pairing)

Adine(A)->thymine(T)

Guanine(G)->Cytosine(C)

Diagram;

Every cell contains the same DNA

DNA releases specific proteins to perform their functions. The proteins can be on or off

Proteins allow cells to act and look differently

An organism inherits genes from parents

Chromosomes are thread like structures

Genes are chains of pairs of DNA bases that vary in length

There are thousands of genes

Genes are different between organisms so that's why we all have different characteristics such as eye colour, hair colour and more.

E.g. Gene 14 could be responsible for hair colour but gene 14 varies between different people.

If a gene is mutated it duplicates, causing cancer

Gametes and inheritance

There types of reproductive cells: sperm cells and egg cells

Egg(ovum) cell:

1. Cytoplasm
2. Cell membrane
3. Jelly coat

Diagram;

Sperm cell:

1. Nucleus
2. Midpiece(contains mitochondria)
3. acrosomes
4. flagellum(tail)

5. Cell membrane
6. Cytoplasm

Diagram:

These cells are considered specialized cells because they have adapted to fit their function

Flagellum: swim to egg

Enzymes in head; releases it break through and digest the through jelly coat for fertilization

Mitochondria: to release energy(respiration) for tail to swim and move to egg

Cytoplasm: Provides energy for zygote to perform mitosis/myosis

Jelly coating that changes after fertilization: to prevent other sperms for penetrating after one has entered

A sperm cell 10,000 smaller than an egg cell

A sperm cell can reach an egg after 15-45 min of ejaculation

Fertilisation: when a sperm cell nucleus fuses with an egg cell nucleus, producing a zygote

Each sex cell has 23 chromosomes in their nucleus, they add up to make 46 when fertilization happens.

The sperm cell either contains and X chromosome or a Y chromosome

An egg cell only contains a X chromosome

If A sperm cell has a Y chromosomes it will be a boy($X+Y=\text{boy}$) and vice versa

Inheritance= passing DNA(as chromosomes with genes) from parents to offspring

The chromosome from your parents will give you a special genetic code

Sexual reproduction needs intercourse for a baby to be produced(animals/humans)

However asexual reproduction does not need intercourse for an offspring to be produced, a mushroom does through mitosis/meiosis

Plants use sexual reproduction by releasing pollen(in flower) that then fuses with an egg also)

A recessive gene is when a gene portrays a trait only when it has 2 identical copies

alleles= A different version of the same gene

genotype= the combination of genes a person has

Homozygous(AA/aa)= when alleles a person has are the same

Phenotype= The physical appearance of someone based on their genotype

heterozygous(Aa)= when two people have 2 different alleles for the same gene. The dominant allele will show its traits

A person inherits a 1 dominant gene and one recessive means that they don't carry the trait of the gene but they can pass on the recessive gene to their offspring.

Even though the recessive gene is not a phenotype in a person it can be a phenotype in their offspring. Because if a this person reproduces with another with a heterozygous, or a recessive dominant genotype then there is a chance for them to produce an offspring with the recessive phenotype

A dominant gene that expresses traits even if it only has one of them.

Examples of dominant genes:

1. Brown eyes
2. Dimples
3. Tongue rolling
4. Freckles
5. Widow's peak

Examples of recessive genes:

1. Attached earlobes
2. Blue eyes
3. Straight hairline
4. Inability to roll a tongue

Capital letter= dominant gene

Lower-case letter= recessive gene

Ff= dominant gets expressed

FF= dominant get expressed

ff= recessive gets expressed

The probability of a child getting the recessive allele or dominant allele can determined using punnet squares

On the top of the squares input parent one's alleles. One on the first column, one on the other. Then input parent two's alleles on the rows. This way we can match the parents' alleles up to see the probability

Diagram:

The probability can be presented in ratios or percentages

Questions will also ask for probability for phenotypes/genotypes (homozygous dominant, homozygous recessive or heterozygous)

Variation

Species = a group of organisms that share the same features, and produce a fertile offspring

Variation = the difference between variations of a species

2 sisters inherit genes from their parents so they are similar not identical) to them

Both genes and your environment influence your characteristics

Genetics:

1. Blood type
2. Eye colour
3. Natural hair colour
4. Earlobe shape

Environment;

1. Hair length
2. Language
3. Scars
4. Tattoos
5. Piercing

Combination of both:

1. Height
2. Skin colour
3. Weight
4. Sporting achievements

Inherited= Characteristics caused by genes found in the nucleus of all cells. Can be passed on from parent to offspring

Environmental= Characteristics caused by influences in the environment. Cannot be passed on from parent to offspring

Both= some characteristics can be both influenced by genes and the environment

A lion's mane, a leopard's spots and a tiger's stripes are all inherited

Two main categories of variation: continuous variation and discontinuous variation

Continuous variation= shows a wide range of intermediate values between two extremes, can be measured

Discontinuous variation= where individuals exhibit distinct, separated categories or groups, with no intermediate values in between

Continuous variation:

1. Mass
2. Height
3. Body length
4. Tail length

5. Ear length
6. Palm size

Discontinuous variation:

1. Eye colour
2. Fur pattern
3. Gender
4. Blood group
5. Fingerprints

Continuous variation can be affected by both environmental and inherited factors

Discontinuous variation can only be affected by inherited factors

Continuous variation: histogram(no space)

Discontinuous variation: bar-chart(space)

Mitosis= when cells split into identical clones

Myosis= when a cell splits to form a distinct cell

Natural Selection

Natural selection= characteristics of organisms that favour its survival, promoting that trait for in future generations

The fitter the organism the higher chance of it surviving and passing on the alleles. This is known as survival of the fittest.

Some species have different variation where some are more resistable to the environment and some are not

An example of this is a peppered moth

Before the industrial revolution, the most peppered moths were pale because they could hide on the bark of the trees.

There was another variation of this moth that was darker, but there were less of those because less of those survived among the trees from predators. This means the “pale” gene is more likely carried on from generation to generation

This is survival of the fittest. The pale moths survived because they could camouflage from the predators.

Then during the industrial revolution, trees turned darker so the more of the darker moths were appearing. This is because more survived, more of them produced offspring, and more of them carried the “dark” gene.

This shows how natural selection relies on the environment and the variation between creatures.

Some bacteria are more resistant than others that are more susceptible to antibacterials. The weaker bacteria die off first then the more resistant survive and reproduce. These offspring carry the same genes so they are more resistant to antibacterial. Over time, most bacteria will be protected from antibacterial so scientists have to find ways to create new medicine to prevent this from happening.

Stages to natural selection

1. Variation
2. There is a lot of variation
3. A condition appears
4. The condition kills off the susceptible organisms
5. The resistant organisms reproduce(due to their variation they survive)
6. The useful, gene that made them survive is passed on
7. Now the condition does not harm the organisms as they all have the ‘resistant’ gene
- 8.

Now doctors urge people to only take anti =boitivs when needed so that the bacteria doesn't get resistant.

Natural selection theory was created by Darwin

He went to island realised some lizards survived the big storm and more of those types of lizards appear due to natural selection

Experiment:

1. Get different types of "beaks"(pick, tweezer, spoon)
2. Try pick up different objects(food)
3. Record how much each "beak" gets
4. Infer which one will survive with natural selection

IV: "beaks

DV: How much the "beaks" picks up of each "food"

CV: amount of time you can try get the food

Use a bar graph to record results and see which variation of bird survives from natural selection

Chemistry: Properties of Materials

Atomic Structure

Atoms are the smallest unit of matter

Atoms are made from subatomic particles

Subatomic particles:

1. Protons
2. Electrons
3. Neutrons

Nucleus is made from proton, neutrons

There shells around the nucleus of an atom

The electrons lie on the shells

Proton: +1

Electron: -1

Neutron: 0

Protons have a mass of one

Electrons have a mass of 0.0005

Neutrons have a mass of 1

Most of the mass is found in the nucleus

1st electron shell: 2 electrons

2nd electron shell: 8 electrons

3rd electron shell: 8 electrons

4th electron shell: 8 electrons

Number of electrons= number of protons

Atomic number= number of protons

Atomic number= Neutron+proton(nucleus)

To find number of neutrons = Atomic mass- atomic number

Valence electrons= the number of electrons on the outer-most shell

An atom overall has no charge as it has the same number of protons as electrons so the net charge=0

An isotope is an element with the same number of protons(atomic number) but have a different number of neutrons(different atomic mass)

E.g Lithium-6(3 neutrons), lithium-8(5 neutrons), Lithium-7(4 neutrons)

All these are isotopes of lithium as they all have the same atomic number but different mass

Only one or two of the element isotopes are stable.

The unstable isotopes decay into other elements by releasing radiation

This is radioactive decay.

The electrons in shells increase in energy as it moves away from the nucleus.

Sometimes if an electron gains enough energy (gets excited) it jumps to a further shells(higher energy levels)

It gains energy through electromagnetic radiation

It will eventually drop back to a lower shell(lower energy levels) and emit the energy in the form of electromagnetic radiation.

To write out the electron structure, write the number of shells(starting from the closest to nucleus) and after each shell put a comma

E.g.: 2,8,8,

Electrostatic attraction hold the atom together

Groups= vertical columns

Periods= horizontal rows